

Held-out test item
Q · source context · reference answer



01

Factor levels

Defines the orthogonal experimental factors and their evaluated levels.

MODEL
Gemma 4
MedGemma 1.5

STATE
 W_0 Base
 $W_0 + BA$ QLoRA

RETRIEVAL
∅ no_rag
≠ hybrid
∧ HyDE



02

Model × adaptation matrix

Crosses each base architecture with its corresponding adaptation state.

	BASE	GROUNDING QLoRA
Gemma 4	W_0	$W_0 + BA$
MedGemma 1.5	W_0	$W_0 + BA$



$$\underbrace{2}_{\text{models}} \times \underbrace{2}_{\text{states}} \times \underbrace{3}_{\text{retrieval modes}} = 12 \text{ conditions}$$

Paired inference
greedy decoding

Provenance
condition + qa_id

RAGAS
[F AR AC SS]